

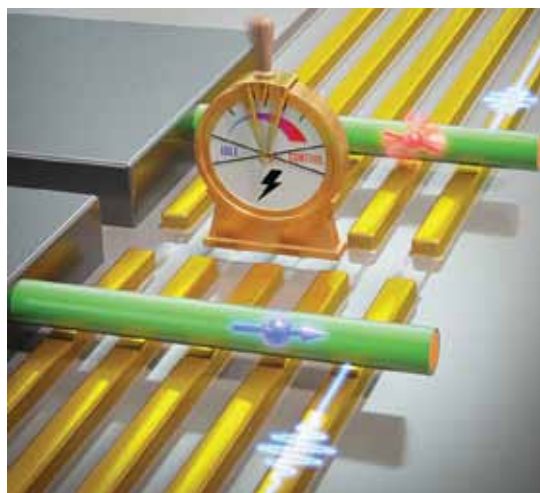
supply power to the drone's rotors and to the controller for the two stacks. Because of the wide range and variable output voltage of the DP30 power pack, from 40 to 74V, the powertrains

were designed to ensure a tightly regulated 48V, 12A output to the rotor motors of the drone, plus a 12V, 8A output to the stack controller board and fans.

## Qubit that can switch between storage and fast calculation modes

To perform calculations, quantum computers need quantum bits (qubits) to act as elementary building blocks that process and store information. Compared to conventional bits, qubits are much more fragile and can lose their information content very quickly. The challenge for quantum computing is therefore to keep the sensitive qubits stable over a prolonged period of time, while at the same time finding ways to perform rapid quantum operations.

Strengthening this is a new type of qubit developed by researchers from the University of Basel and TU Eindhoven that can be switched from a stable idle mode to a fast calculation mode and store quantum information with the application of an electrical voltage. This capability is vital since the construction of a powerful quantum computer requires the ability to selectively control and interconnect many individual qubits. Scalability is particularly necessary to reduce the error rate in quantum calculations. The researchers were also able to use the electrical switch to manipulate the spin qubits at record speed in as little as a nanosecond.



Electrically switchable qubit: a nanowire made of germanium and silicon (blue/green) lies on electrodes known as gates (gold). Voltages applied to the gates lead to the formation of individual spin qubits (blue and red arrows) that can be manipulated by microwave signals (blue pulse). In one mode, the qubit is slow and the quantum information is more stable (blue spin). In the other, the qubit can be changed more quickly (red spin) (Credit: [www.unibas.ch/en](http://www.unibas.ch/en))

## Neuroscience and robotics combination helps rehabilitate patients



Gordon Cheng, Professor for Cognitive Systems, wants to dig deeper in understanding how the brain works (Credit: [www.tum.de/nc/en](http://www.tum.de/nc/en))

Combining neuroscience with robotic research has gained impressive results in the rehabilitation of paraplegic patients. A research team from the Technical University of Munich (TUM) was able to show that exoskeleton training not only helped patients to walk but also stimulated their healing process, allowing them to regain a certain degree of control over the movement of their legs.

The idea behind this is that the coupling between the brain and the machine should work in a way where the brain thinks of the machine as an extension of the body. Given the recent technological advances and better understanding of how to decode the user's momentary brain activity, the time is ripe for their integration into more human-centered solutions.

## Smart devices and AI help monitor mental health

Behavioural health issues like depression and bipolar disorder often do not clearly show up with external symptoms and cannot be easily detected (unlike common cold). While there are thousands of apps available to assist in mental wellbeing and mental-health monitoring, these tools are rarely validated by clinicians, nor do they deliver behavioural data in ways that make sense to doctors.



(Credit: <https://infosci.cornell.edu>)